

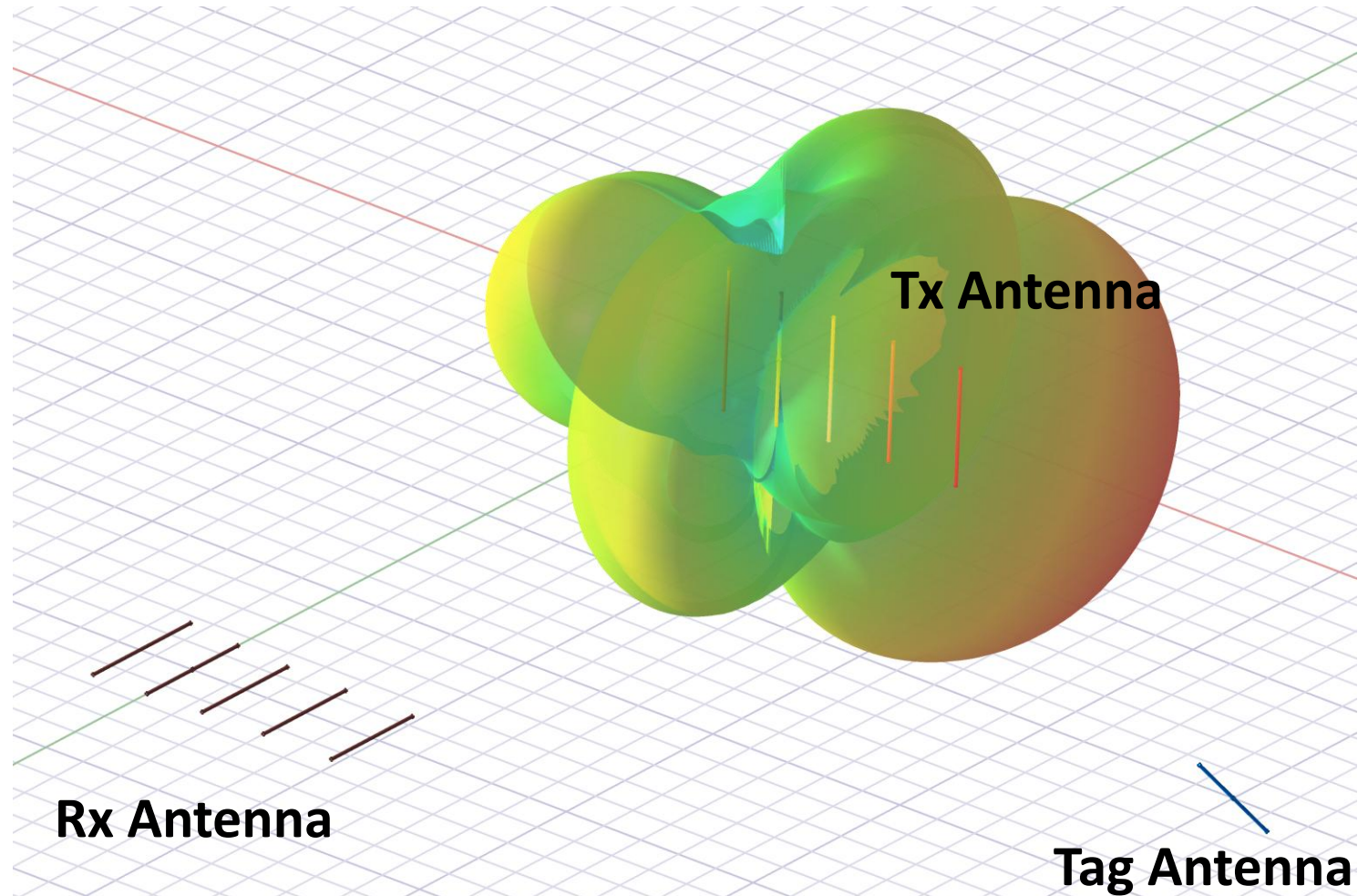
1 Bistatic Topology CAD Model

1. Goal: Verify antenna placement, orientation, and geometry before running numeric simulations.

2. Technical Description:

- Tx Yagi at origin; Rx Yagi offset along X-axis (bistatic baseline)
- Two 5-element Yagi arrays: reflector, driven dipole, 3 directors
- Tag dipole placed off-axis, oriented 45° in Y-Z plane
- ~170 mm element lengths for 866 MHz

3. Result: The 3D model geometry, spacing, and orientation match the intended bistatic cross-pol design.



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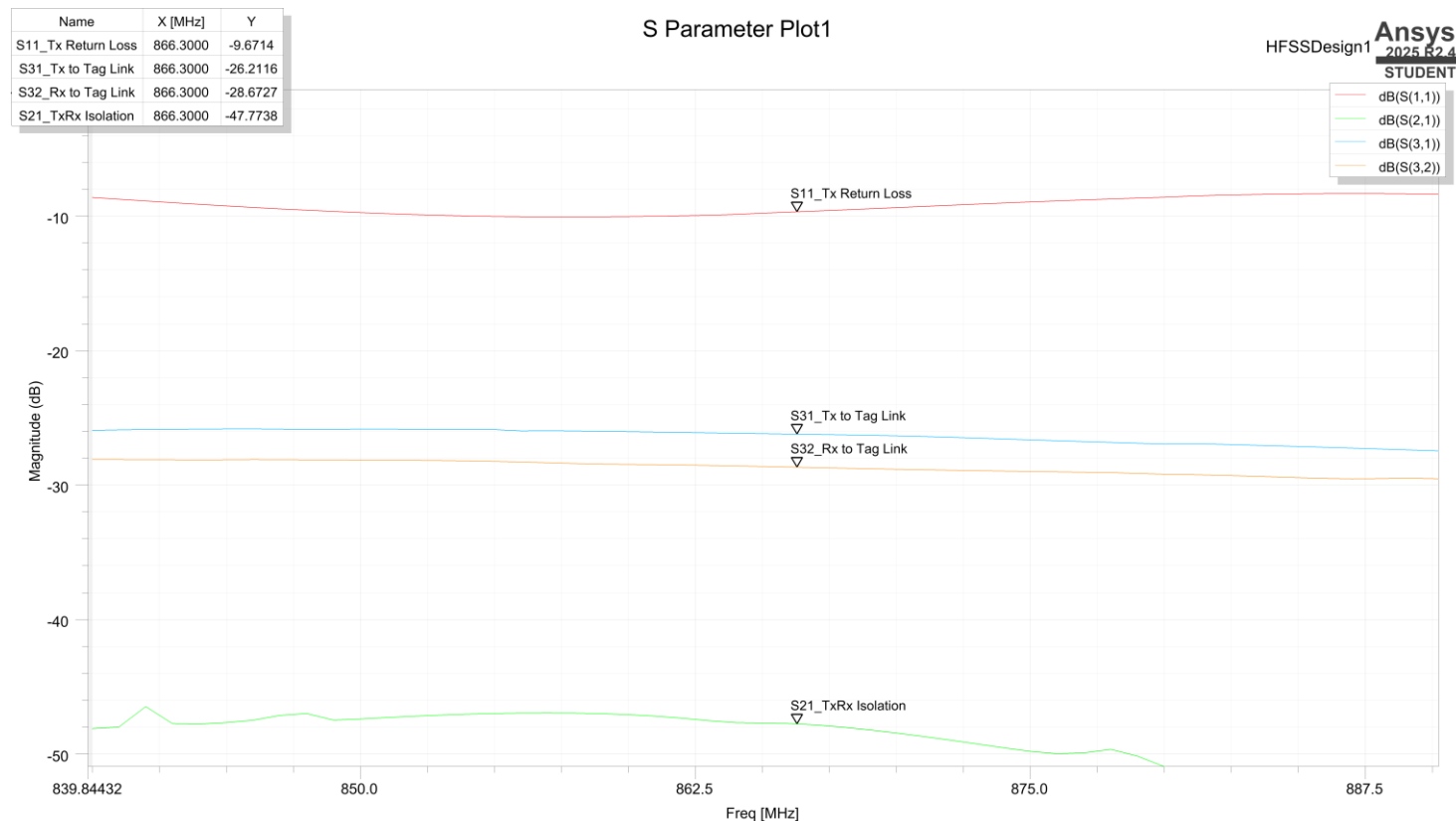
Cross-Polarization Isolation Sweep

1. Goal: Quantify impedance match and Tx-to-Rx isolation across the operating band.

2. Technical Description:

- Swept 839.8–887.5 MHz, centered at 866.3 MHz (India SRD band)
- $S_{11} = -9.67$ dB $\rightarrow$ VSWR $\approx 1.98:1$
- $S_{31} = -26.21$ dB (Tx $\rightarrow$ Tag)
- $S_{32} = -28.67$ dB (Rx $\rightarrow$ Tag)
- $S_{21} = -47.77$ dB (Tx $\rightarrow$ Rx direct leakage)

3. Result: Direct leakage isolation (-47.77 dB) beats the -30 dB target, validating the cross-polarization design.



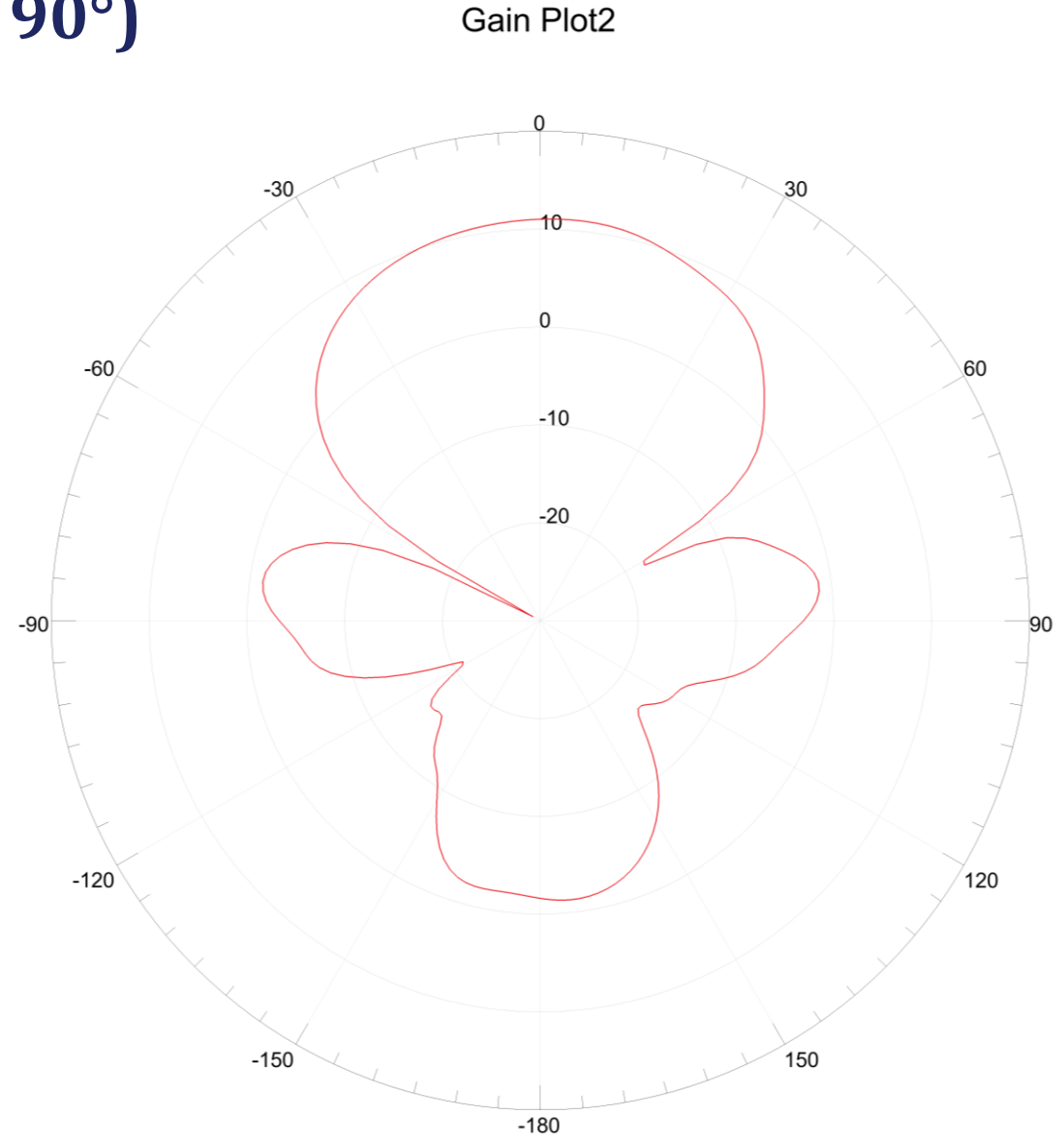
3 Tx Azimuth Cut — H-Plane ($\theta = 90^\circ$)

1. Goal: Characterize the Tx antenna's horizontal-plane radiation pattern.

2. Technical Description:

- Cut taken at Theta = 90° , 866.3 MHz
- Pattern shows four lobes instead of one clean lobe
- Extra lobes caused by Rx Yagi and Tag dipole scattering nearby

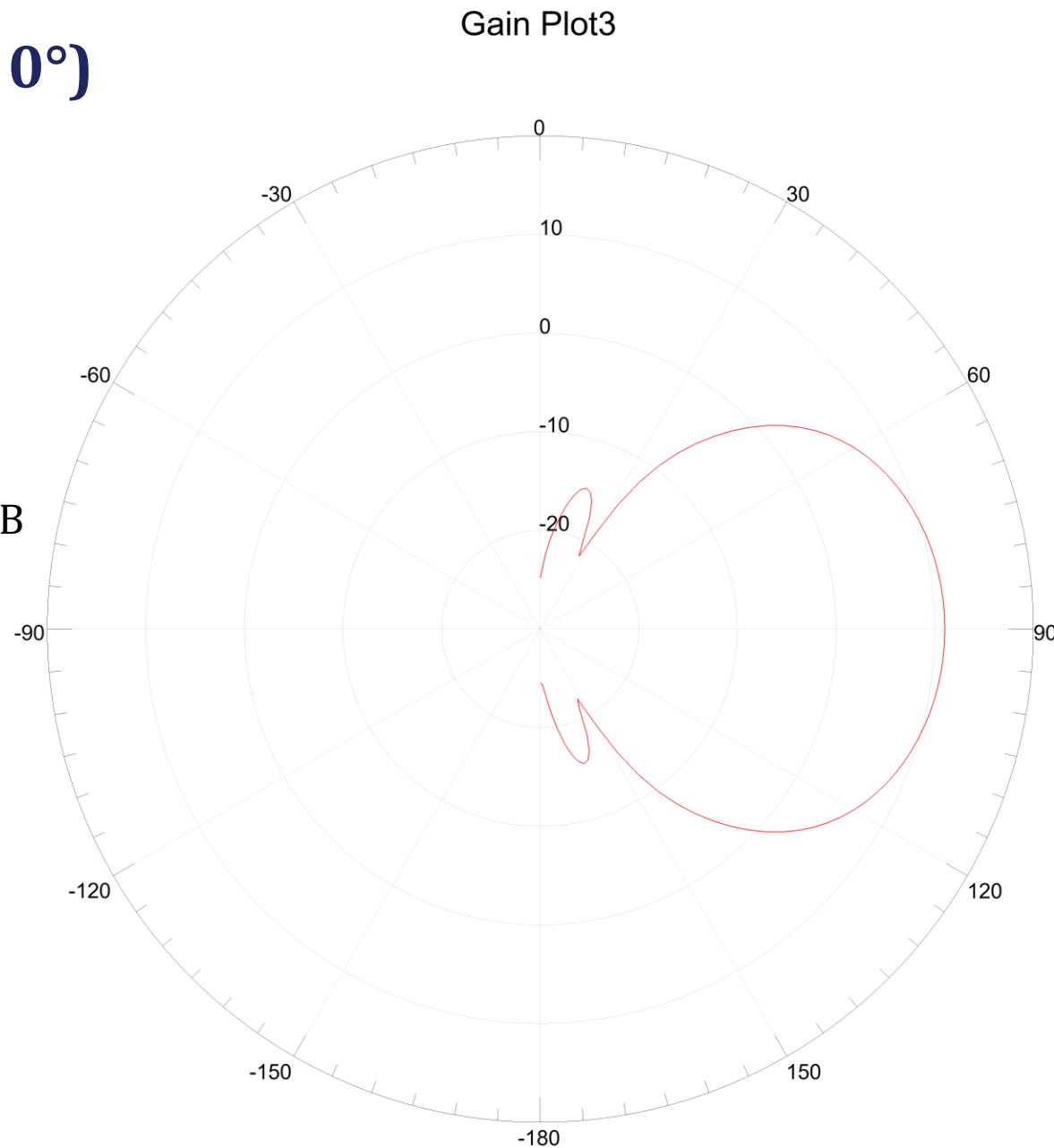
3. Result: The distortion is real mutual coupling from nearby antennas, not a simulation error.



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Tx Elevation Cut — E-Plane ($\varphi = 0^\circ$)

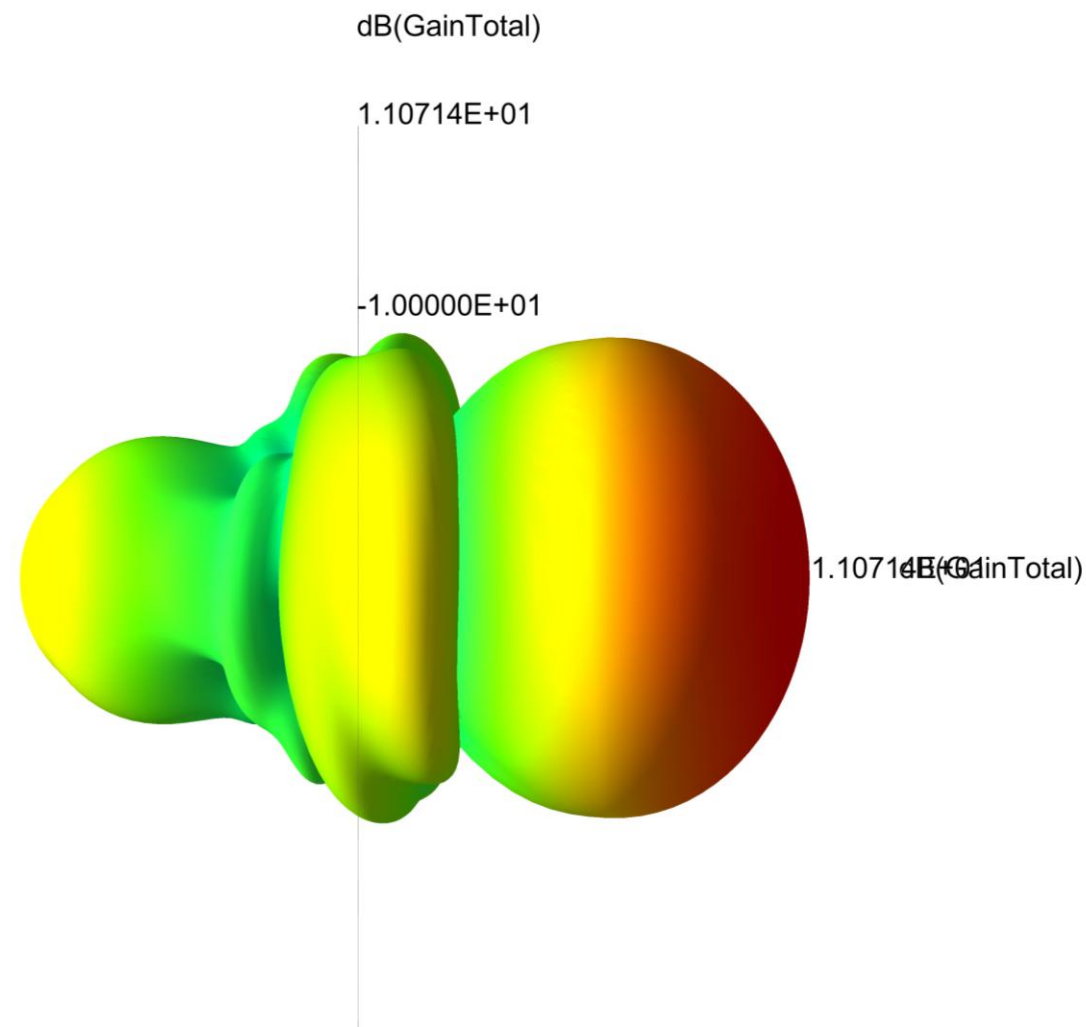
- 1. Goal:** Characterize the Tx antenna's vertical-plane radiation pattern.
- 2. Technical Description:**
 - Cut taken at $\Phi = 0^\circ$, 866.3 MHz
 - Single clean main lobe (~ 10 dB) toward +X
 - Back lobe ~ -20 dB $\rightarrow$ front-to-back ratio ≈ 30 dB
- 3. What was proved:** The Tx Yagi delivers textbook directivity matching its 12 dBi datasheet rating.



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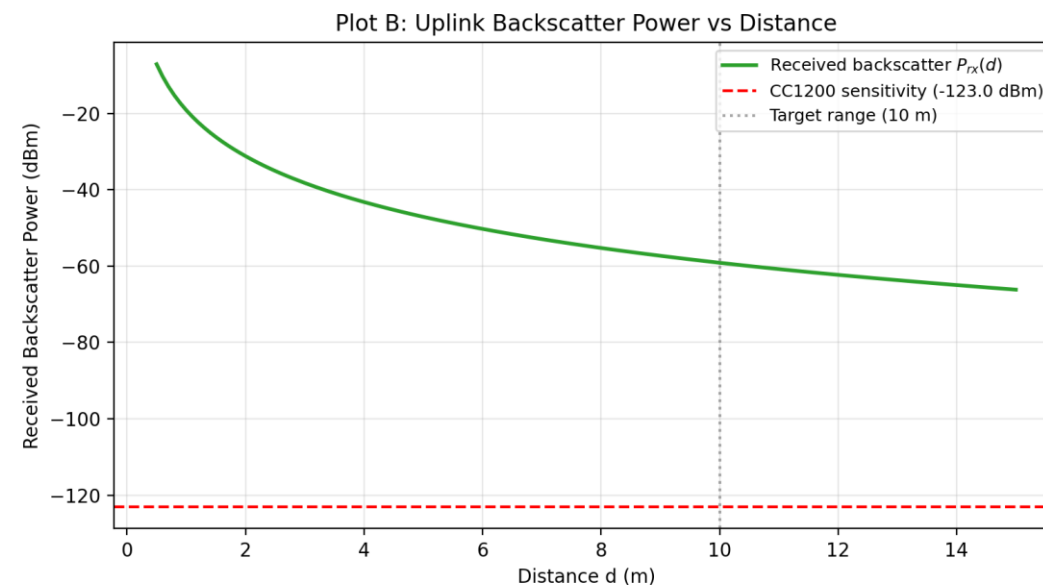
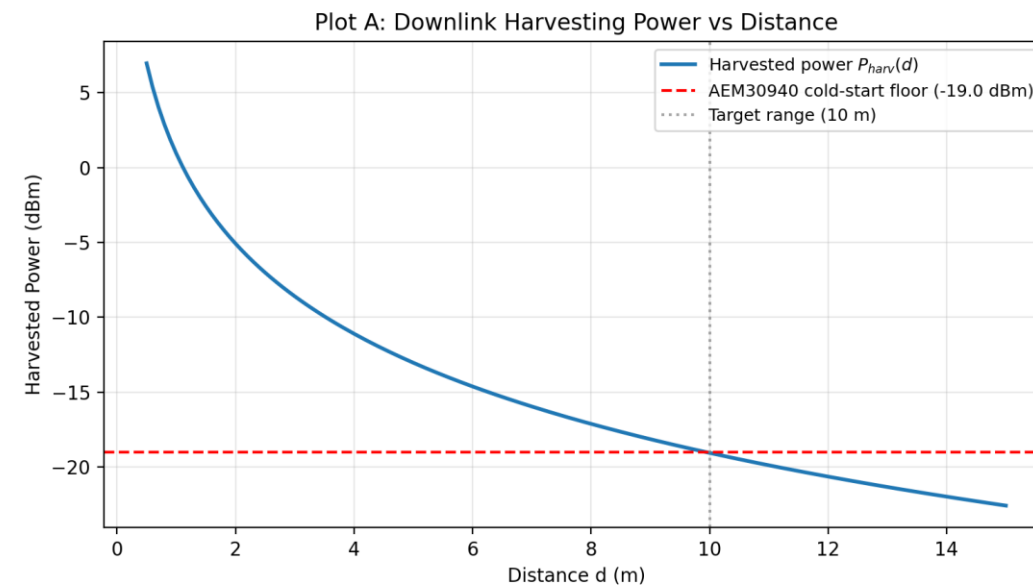
3D Far-Field Realized Gain Pattern

- 1. Goal:** Confirm overall 3D gain performance and directional behavior of the Tx antenna.
- 2. Technical Description:**
 - Peak gain = 11.07 dB, Min = -38.70 dB
 - Small ~ 1 dB shortfall from conductor loss/meshing
 - Strong forward lobe (+X), weak rear lobe
- 3. Result:** The antenna's realized gain and directivity closely match design expectations.



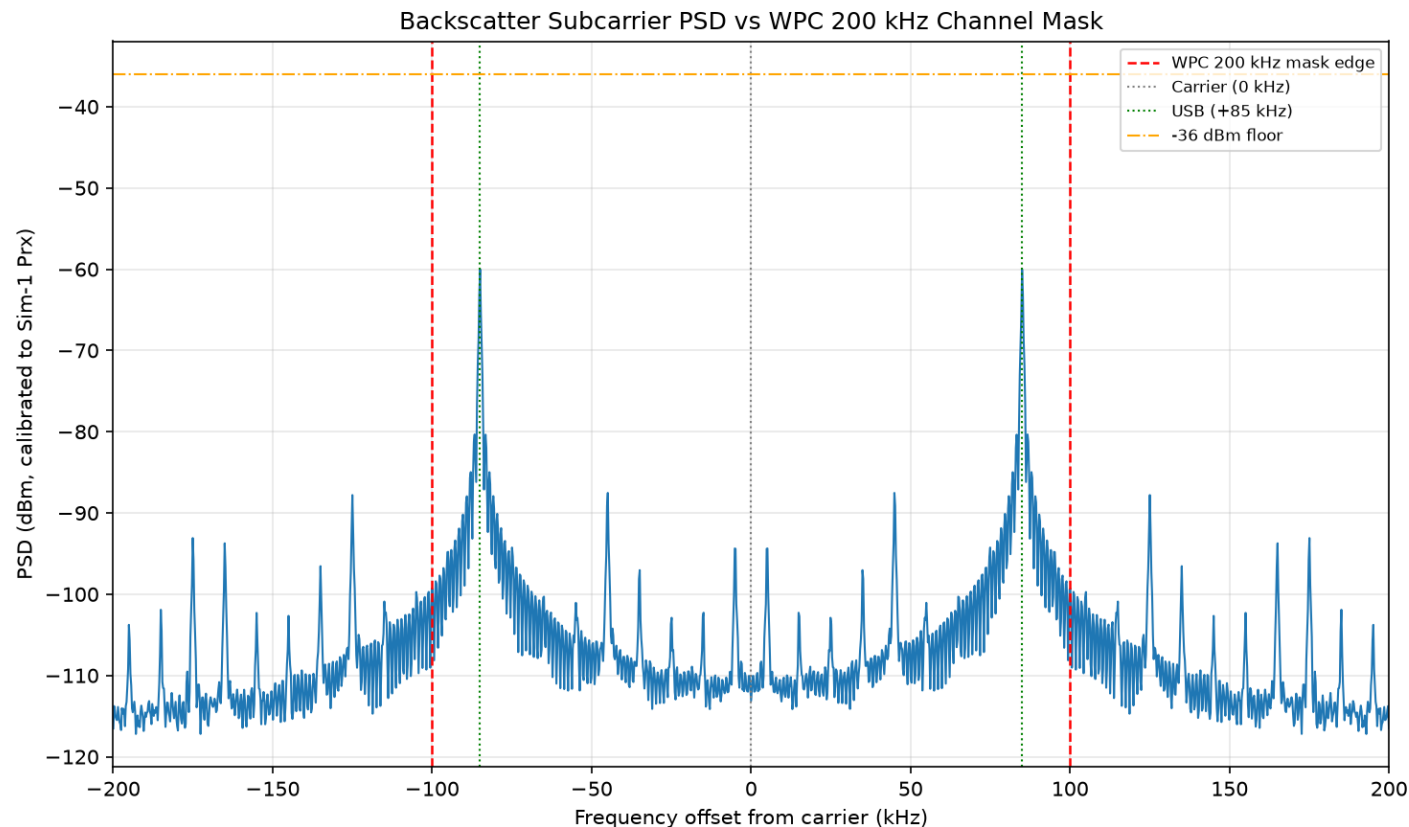
6 Bistatic Radar Range & RF Harvesting Link Budget

- Goal:** Prove the batteryless node can cold-start and backscatter reliably out to a 10 m operating range under the WPC EIRP limit.
- Technical Description:**
 - Friis one-way path used for downlink harvesting; two-way radar equation used for uplink backscatter
 - Tx/Rx Yagi gains = +12 dBi each, tag dipole gain = +2.15 dBi, 45° polarization mismatch loss applied
 - Swept distance 0.5–15 m against AEM30940 cold-start floor (−19 dBm) and CC1200 sensitivity (−123 dBm)
- Result:** Harvested power crosses the −19 dBm cold-start floor almost exactly at 10 m (near-zero margin), while backscatter power at 10 m is still ∼−60 dBm — over 60 dB above the CC1200 sensitivity floor. Range is bottlenecked by the harvesting link, not the backscatter link.



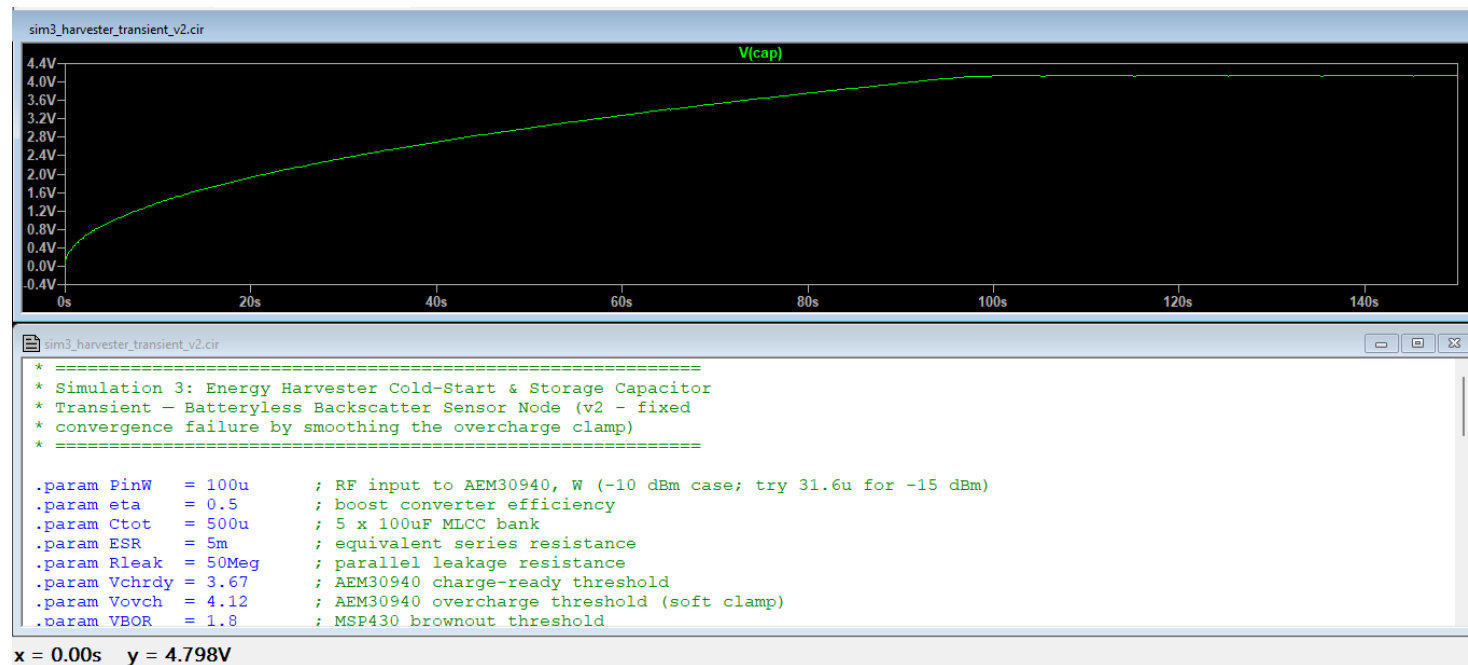
WPC Regulatory Subcarrier Spectrum & Bandwidth Compliance

- 1. Goal:** Confirm the 85 kHz OOK subcarrier stays within the WPC 200 kHz channel mask with adequate guard band.
- 2. Technical Description:**
 - Square-wave 85 kHz subcarrier OOK-modulated with PRBS data, processed via Welch PSD
 - Mask boundaries set at ± 100 kHz, regulatory floor set at -36 dBm
 - USB/LSB tones and harmonics checked against both boundaries
- 3. Result:** USB/LSB peaks land at ± 85 kHz (≈ -60 dBm), sitting inside the ± 100 kHz mask with the required ≥ 15 kHz guard band; harmonics beyond ± 100 kHz drop below -90 dBm, well under the -36 dBm floor — spectrum is compliant.



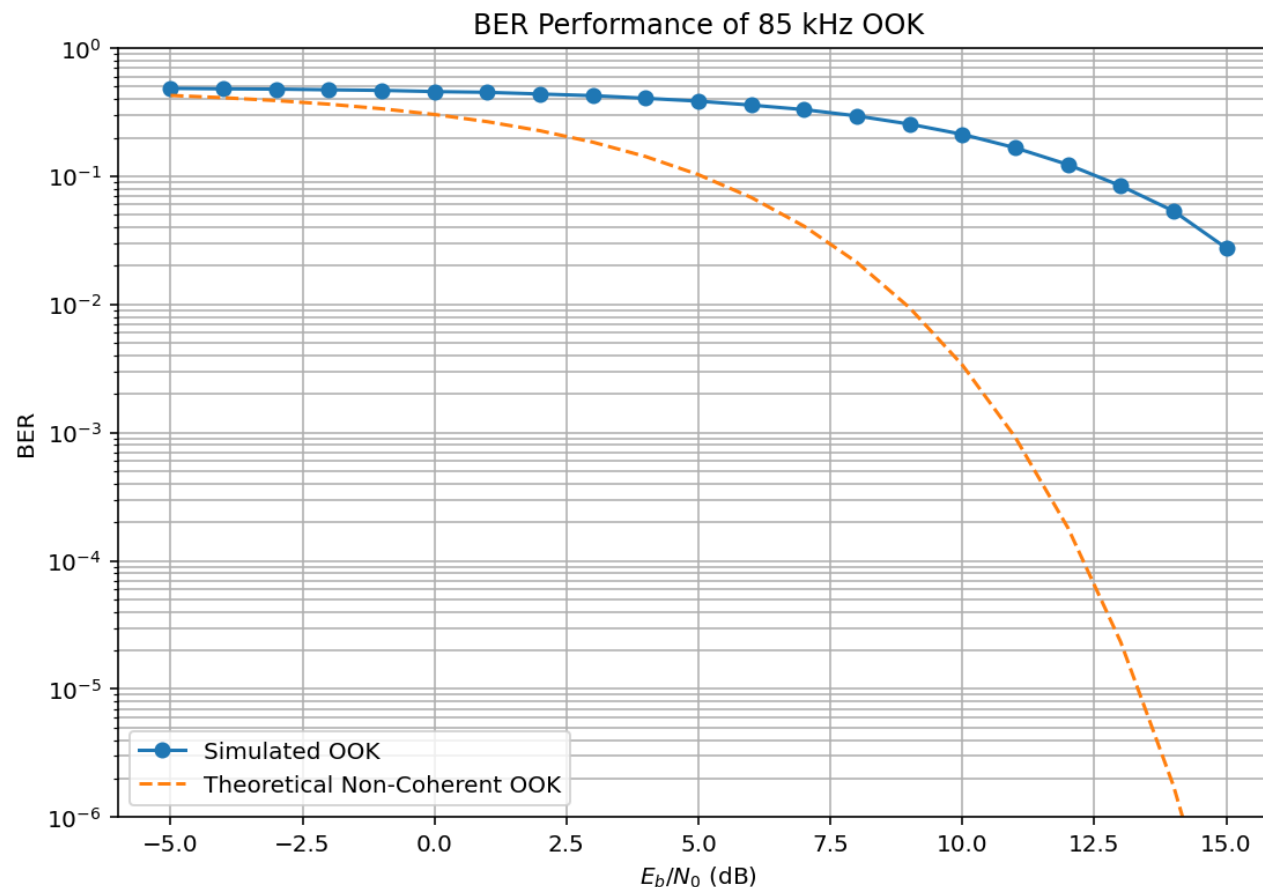
Energy Harvester Cold-Start & Storage Capacitor Transient

1. **Goal:** Validate that the 500 μF storage bank can cold-start from 0 V to the AEM30940 charge-ready threshold without instability.
2. **Technical Description:**
 - Behavioral current source models the AEM30940 boost stage ($\eta \approx 50\%$, 100 μW RF input)
 - 5 $\times$ 100 μF MLCC bank with ESR = 5 m Ω and Rleak = 50 M Ω
 - Transient (.tran) run captures the full charge ramp; clamp smoothing fixed an earlier convergence failure
3. **Result:** V(cap) rises smoothly from 0 V, crosses the 3.67 V charge-ready threshold, and settles near the ~ 4.1 V overcharge clamp without oscillation — confirming a stable, convergence-safe cold-start.



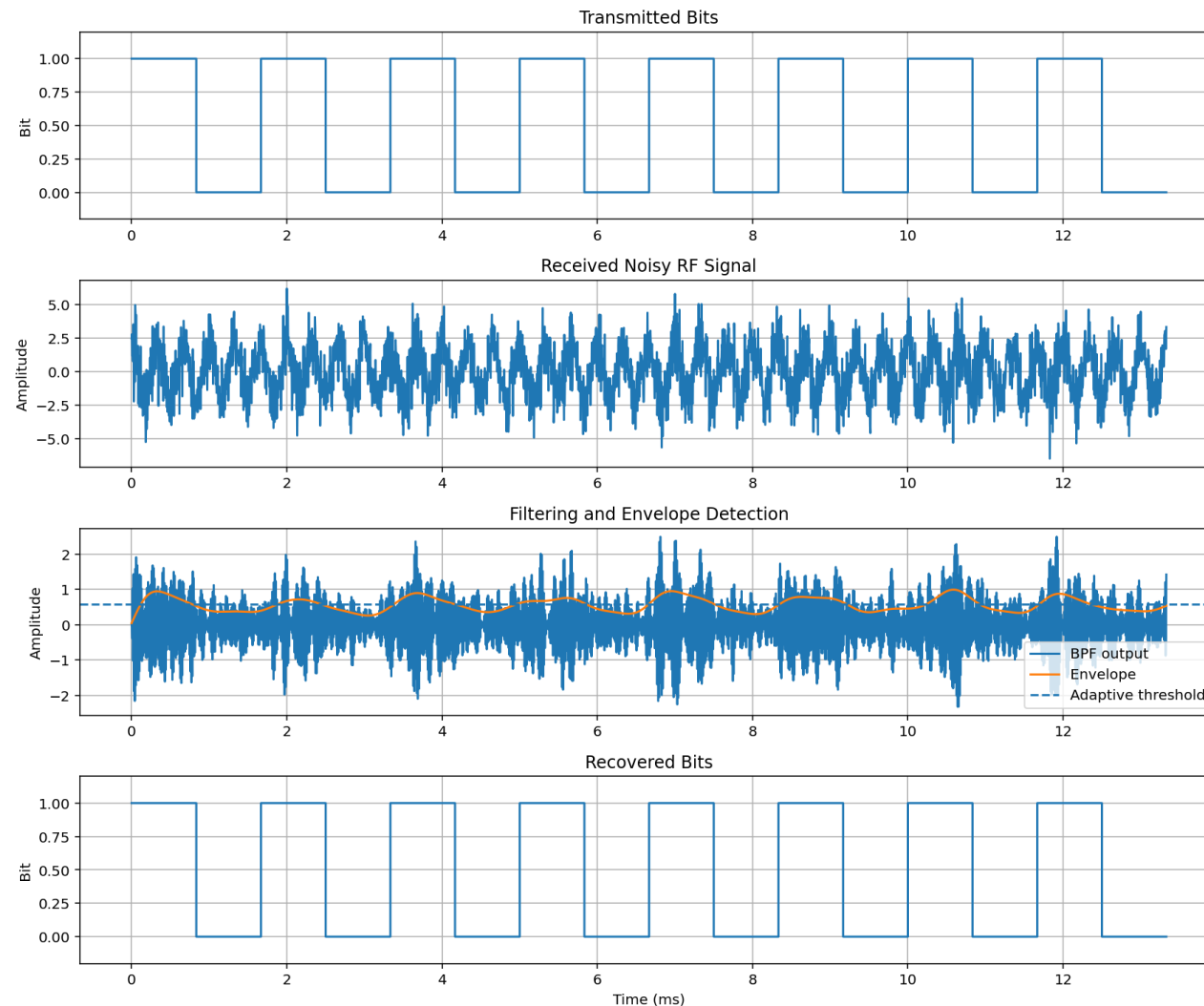
9 Baseband Subcarrier BER Performance (Waterfall Curve)

1. **Goal:** Determine the minimum E_b/N_0 needed for the demodulator to hit the <1% PER target.
2. **Technical Description:**
 - Full Tx → channel → BPF → envelope-detector → slicer pipeline simulated via Monte Carlo
 - AWGN swept from -5 dB to +15 dB E_b/N_0 , compared against theoretical non-coherent OOK
 - BER computed per step with sufficient bit count for statistical validity
3. **Result:** Simulated BER lags the theoretical curve significantly, flattening near ~3% at 15 dB E_b/N_0 instead of following the theoretical curve down to $\sim 10^{-6}$ — indicating real implementation losses and that the <1% PER target is not yet met in the tested SNR range.



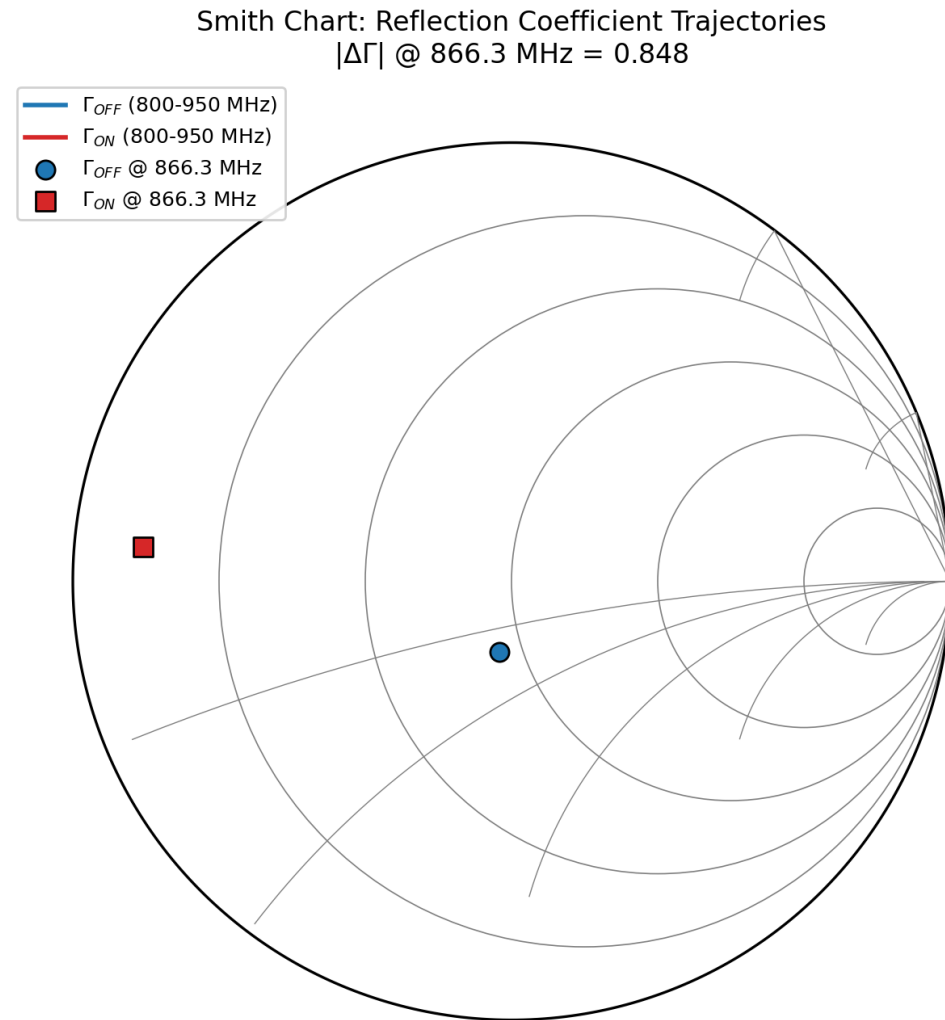
Baseband Demodulation Signal Stages (Time-Domain Pipeline)

1. **Goal:** Visually verify each stage of the receiver chain correctly recovers the transmitted bitstream.
2. **Technical Description:**
 - Panel 1: clean transmitted OOK bit sequence
 - Panel 2: received signal fully buried in noise/carrier leakage, bit edges not visible
 - Panel 3: bandpass-filtered + envelope-detected signal with adaptive threshold overlaid
 - Panel 4: final recovered bitstream
3. **Result:** At this operating SNR, the recovered bitstream matches the transmitted sequence, confirming the BPF → envelope-detection → threshold-slicer chain works end-to-end — though the BER curve shows this success rate degrades at lower SNR.



Shunt Modulator S-Parameters & Reflection Contrast (Smith Chart)

1. **Goal:** Visualize the ADG902 switch's reflection coefficient trajectory in OFF vs ON states to confirm strong modulation contrast.
2. **Technical Description:**
 - Γ_{OFF} and Γ_{ON} plotted across 800–950 MHz sweep
 - Markers highlight exact reflection coefficient at 866.3 MHz for both states
 - $|\Delta\Gamma|$ computed as the vector distance between Γ_{OFF} and Γ_{ON} on the chart
3. **Result:** Γ_{OFF} sits near the chart center ($\approx$ matched, minimal reflection), while Γ_{ON} is pulled far toward the left boundary (near short-circuit). Measured $|\Delta\Gamma| = 0.848$ at 866.3 MHz — a strong contrast, though slightly short of the ideal unity ($|\Delta\Gamma| \approx 1.0$) due to the switch's non-zero R_{on} (4.5 Ω) and package inductance.



Shunt Modulator S-Parameters (Cartesian: Insertion Loss & Reflection)

- Goal:** Quantify insertion loss in harvester mode (OFF) and reflection strength in backscatter mode (ON) across the band.
- Technical Description:**
 - Top: S21 vs frequency in State 0 (OFF) — harvester through-path loss, compared to a -0.8 dB target
 - Bottom: S11 vs frequency in State 1 (ON) — backscatter reflection magnitude, compared to a -0.5 dB target
 - Both swept 800–950 MHz with 866.3 MHz marker
- Result:** S21 (OFF) stays between -0.10 and -0.15 dB across the band — well under the -0.8 dB limit, confirming minimal harvesting loss. S11 (ON) sits around -1.44 to -1.49 dB, falling short of the -0.5 dB strong-reflection target ($|\Gamma| \approx 0.85$ vs. ideal ≈ 1.0), consistent with the $|\Delta\Gamma| = 0.848$ seen on the Smith chart — the switch delivers low insertion loss but slightly reduced reflection contrast versus the ideal case.

